Coupled 1st Order Linear DEs

Questions 1–30

Question 1. It is given that $\frac{dy}{dt} = y$.

a) Solve this differential equation to find a general solution for y.

It is also given that $\frac{dx}{dt} = x + y$.

b) Using your result from part (a), solve this second differential equation to find a general solution for x.

Question 2. It is given that $\frac{dx}{dt} = 2x$.

a) Solve this differential equation to find a general solution for x.

It is also given that $\frac{dy}{dt} = x + 2y$.

b) Using your result from part (a), solve this second differential equation to find a general solution for y.

Question 3. It is given that $\frac{dx}{dt} = x + y$ and $\frac{dy}{dt} = 2y$.

a) Rearrange the first equation to express y in terms of x and $\frac{dx}{dt}$.

b) Hence, find an expression for $\frac{dy}{dt}$ in terms of x, $\frac{dx}{dt}$ and $\frac{d^2x}{dt^2}$.

c) By substituting your expressions from parts (a) and (b) into the second equation, show that $\frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 0$.

d) Solve this differential equation to find a general solution for x.

e) Hence find a general solution for y in terms of t.

Question 4. It is given that $\frac{dx}{dt} = 2x + y$ and $\frac{dy}{dt} = x + 2y$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find a general solution for x in terms of t.

c) Hence find a general solution for y in terms of t.

Question 5. It is given that $\frac{dx}{dt} = x + y$ and $\frac{dy}{dt} = -4x + 5y$.

a) Form a second-order differential equation in terms of y.
b) Solve this differential equation to find a general solution for y in terms of t.
c) Hence find a general solution for x in terms of t.

Question 6. It is given that $\frac{dx}{dt} = y$ and $\frac{dy}{dt} = -x + 2y$.

a) Form a second-order differential equation in terms of x.
b) Solve this differential equation to find the general solution for x in terms of t.
c) Hence find the general solution for y in terms of t.

Question 7. It is given that $\frac{dx}{dt} = x + 3y$ and $\frac{dy}{dt} = x - y$.

a) Form a second-order differential equation in terms of y.

b) Solve this differential equation to find the general solution for y in terms of t.

c) Hence find the general solution for x in terms of t.

It is also given that when $t = 0$, $x = 4$ and $y = 0$.

d) Find the particular solutions for x and y.

Question 8. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\frac{dx}{dt} = 2x + y, \quad \frac{dy}{dt} = x + 2y$

b) $\frac{dx}{dt} = 2x + 2y, \quad \frac{dy}{dt} = -2x + 6y$

c) $\frac{dx}{dt} = x + 2y, \quad \frac{dy}{dt} = -x - y$

d) $\frac{dx}{dt} = 2x + y, \quad \frac{dy}{dt} = -5x$

Question 9. It is given that $\frac{dx}{dt} = x + y + 1$ and $\frac{dy}{dt} = x + y - 1$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find the general solution for x in terms of t.

c) Hence find the general solution for y in terms of t.

It is also given that when $t = 0$, $x = 1$ and $y = 0$.

d) Find the particular solutions for x and y.

Question 10. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\frac{dx}{dt} = 2y + 6, \quad \frac{dy}{dt} = 2x - 4$

b) $\frac{dx}{dt} = 3x + y + 2, \quad \frac{dy}{dt} = -x + y - 2$

c) $\frac{dx}{dt} = 2x + y + 3, \quad \frac{dy}{dt} = -13x - 2y - 5$

d) $\frac{dx}{dt} = 3x + 2y + 1, \quad \frac{dy}{dt} = -2x + 3y + 1$

Question 11. A system of differential equations is given by $\frac{dx}{dt} = x + y + t$ and $\frac{dy}{dt} = 2y + t^2$. It is also given that when $t = 0$, $x = \frac{7}{4}$ and $y = \frac{1}{4}$. Find expressions for x and y in terms of t.

Question 12. It is given that $\frac{dx}{dt} = 2x + y + \sin t$ and $\frac{dy}{dt} = -x + \cos t$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find the general solution for x in terms of t.

c) Hence find the general solution for y in terms of t.

It is also given that when $t = 0$, $x = 1$ and $y = 1$.

d) Find the particular solutions for x and y.

Question 13. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\dfrac{dx}{dt} = 2x + y + \sin t, \quad \dfrac{dy}{dt} = 2x + 3y + \cos t$

b) $\dfrac{dx}{dt} = x + y + \sin 4t, \quad \dfrac{dy}{dt} = -x + 3y + 3\sin 4t$

c) $\dfrac{dx}{dt} = 3x + 2y + \cos 2t, \quad \dfrac{dy}{dt} = -9x - 3y - 5\cos 2t$

d) $\dfrac{dx}{dt} = 2x + y + \sin 5t, \quad \dfrac{dy}{dt} = -x + 2y - 25\cos 5t + 22\sin 5t$

Question 14. A system of differential equations is given by $\dfrac{dx}{dt} = 2x + 3y + e^{-2t}$ and $\dfrac{dy}{dt} = x + 2y + e^{-2t}$. It is also given that $x(0) = \dfrac{25}{13}$ and $y(0) = -\dfrac{3}{13}$. Find expressions for x and y in terms of t.

Question 15. A system of differential equations is given by $\frac{dx}{dt} = -3x + 4y + e^{-t}$ and $\frac{dy}{dt} = \frac{3}{2}x - 4y + e^{-t}$. It is also given that $x(0) = 1$ and $y(0) = \frac{1}{4}$. Find expressions for x and y in terms of t. **Modelling: Predator Prey**

Question 16. On a nature reserve, rabbits and foxes interact. At time t years, x is the number of rabbits, and y is the number of foxes. The populations follow $\frac{dx}{dt} = 0.3x - 12y$ and $\frac{dy}{dt} = 0.02x - 1.5$.

a) Explain the meaning of the $0.3x$ and $-12y$ terms in the equation for $\frac{dx}{dt}$.

b) Explain the meaning of the $0.02x$ and -1.5 terms in the equation for $\frac{dy}{dt}$.

c) State what would happen to the rabbit population if there were no foxes present.

d) State what would happen to the fox population if there were no rabbits present.

e) Explain why the term -1.5, rather than $-1.5y$, might be unrealistic in a real biological model.

Question 17. In a lake, fish are preyed upon by birds. At time t years, x is the number of fish, and y is the number of birds. The system is modelled by $\frac{dx}{dt} = 0.5x - 8y$ and $\frac{dy}{dt} = 0.02x - 1.6$.

a) State the meaning of the constants 0.5, -8, 0.02, and -1.6.

b) Find the equilibrium point (where both populations are constant).

Question 18. On a tropical island, a population of insects has grown rapidly. To control them, frogs have been introduced. At time t years, where $t \geq 0$, x is the number of insects, and y is the number of frogs. Initially, there are 1200 insects and 25 frogs. For $t \geq 0$, it is assumed that:

- Insects reproduce at a rate of p% per year.
- Each frog eats, on average, q insects per year.
- Frogs die at a constant rate of r frogs per year.
- The frog population increases at a rate of s% of the insect population per year.

This leads to the system $\frac{dx}{dt} = 0.4x - 4y$ and $\frac{dy}{dt} = 0.02x - 2$.

a) State the values of p, q, r, and s.

b) Solve the coupled differential equations to find expressions for x and y in terms of t.

c) Given that $x > 0$ and $y > 0$ for $0 \leq t \leq 4$, use your solution to determine whether the insect population is being successfully controlled.

Question 19. In a woodland ecosystem, voles are preyed upon by owls. At time t years, x is the number of voles and y is the number of owls. It is known that the populations can be modelled by equations of the form $\frac{dx}{dt} = Ax + By$ and $\frac{dy}{dt} = Cx + Dy$. For $t > 0$, the following assumptions are made:

- In the absence of predators, the vole population decreases by 5% per year due to limited resources.
- Each owl causes the vole population to decrease by 2 voles per year.
- The owl population increases at a rate of 0.02 times the vole population per year.
- In the absence of food, the owl population decreases by 5% per year.

Initially, there are 300 voles and 20 owls.

a) State the values of A, B, C and D.

b) Explain what it means, in this model, for the owl death term to be proportional to y.

c) Solve the coupled differential equations to find expressions for x and y in terms of t.

d) Find the first time at which the vole population becomes zero. Explain why the model is no longer valid after this time.

Modelling: Two Tanks

Question 20. Tank 1 contains 500 litres of liquid. Salt solution with concentration 4 grams per litre flows into tank 1 at 8 litres per minute. The mixture in tank 1 is well stirred and flows out at 8 litres per minute. Let x be the amount of salt (in grams) in tank 1 at time t minutes.

a) Write an expression for the concentration of salt in the tank.

b) Find the rate at which salt leaves the tank.

c) Show that $\frac{dx}{dt} = 32 - \frac{2}{125}x$.

The liquid leaving tank 1 flows into tank 2, which also contains 500 litres of liquid. In addition, salt solution of concentration 10 grams per litre flows directly into tank 2 at 2 litres per minute. The mixture in tank 2 is well stirred and flows out of tank 2 at 10 litres per minute. Let y be the amount of salt in tank 2 at time t minutes.

d) Write down a differential equation for $\frac{dy}{dt}$ in terms of x and y.

Question 21. Two tanks, A and B, each contain 400 litres of liquid. Liquid flows from A to B at 12 litres per minute. Liquid leaves B at 12 litres per minute. Initially, tank A contains 60 grams of salt and tank B contains none. The liquid in each tank is well-stirred at all times. Let x be the amount of salt (in grams) in tank A at time t minutes, and let y be the amount of salt (in grams) in tank B at time t minutes.

a) Write down an expression for $\dfrac{dx}{dt}$.

b) Show that $\dfrac{dy}{dt} = \dfrac{3}{100}x - \dfrac{3}{100}y$.

c) Solve the coupled differential equations to find expressions for x and y in terms of t.

Question 22. Two tanks, P and Q, each contain 700 litres of liquid. Liquid flows from P to Q at 15 litres per minute, and liquid flows from Q to P at 7 litres per minute. No liquid flows into P or Q from outside. Liquid leaves Q at a rate that keeps volume constant. Let x and y be the amounts of salt in P and Q.

a) Explain why liquid must leave Q at 8 litres per minute.

b) Write down expressions for $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

Question 23. Two tanks, M and N, each contain 900 litres of liquid. Salt solution of concentration c grams per litre flows into M at 7 litres per minute. Liquid flows from M to N at 16 litres per minute, and liquid flows from N to M at 9 litres per minute. Liquid leaves N so that volumes remain constant. Let x and y be the grams of salt in M and N.

a) Show that $\frac{dx}{dt} = 7c - \frac{4}{225}x + \frac{1}{100}y$.

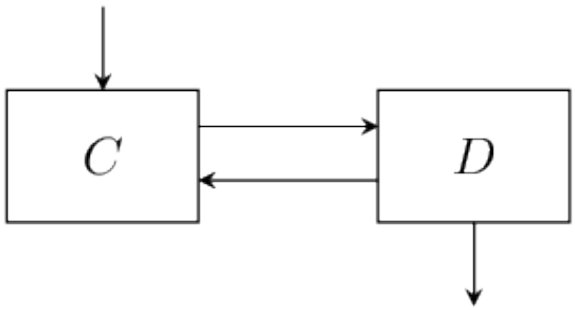
b) Write down a differential equation for y.

c) Given that the system reaches a steady state where $x = 450$ and $y = 450$, find the value of c.

It is given that initially there are 500 grams of salt in tank M and 400 grams of salt in tank N.

d) Solve the coupled differential equations to find expressions for x and y in terms of t.

Question 24. Two large containers, C and D, are each filled with 1000 litres of liquid. At time $t = 0$, both containers contain pure water.



For $t > 0$, the system operates as follows:

- A solution containing p grams per litre of salt flows into container C at a constant rate.
- Liquid flows from container C to container D at 25 litres per minute.
- Liquid flows from container D to container C at 9 litres per minute.
- Liquid leaves container D through an outlet.
- The volume of liquid in each container remains constant.

Let $x(t)$ be the mass of salt, in grams, in container C, and let $y(t)$ be the mass of salt, in grams, in container D.

a) Show that the rate at which liquid flows into container C is 16 litres per minute.

b) Show that the system can be modelled by $\frac{dx}{dt} = 16p - 0.025x + 0.009y$ and $\frac{dy}{dt} = 0.025x - 0.025y$.

c) Given that $p = 4$, solve the system of differential equations to find expressions for $x(t)$ and $y(t)$.

d) Find the limiting values of x and y as $t \to \infty$, and interpret your answer in terms of the concentrations in the two containers.

Modelling: Mix

Question 25. Two substances, A and B, are involved in a reaction inside a closed vessel. Let $a(t)$ and $b(t)$ denote the masses (in grams) of substances A and B respectively, at time t minutes after the reaction begins. The system is modelled by the differential equations $\frac{da}{dt} = -3a + 4b + 6$ and $\frac{db}{dt} = -2a + 2b + 10$.

a) Show that a satisfies $\frac{d^2a}{dt^2} + \frac{da}{dt} + 2a = 28$.

b) Solve this differential equation to obtain the general solution for $a(t)$.

c) Hence find an expression for $b(t)$.

Initially, A has a mass of 4 grams and B has a mass of 1 gram.

d) Find the particular solutions for both $a(t)$ and $b(t)$.

e) Show that the system approaches a steady state as $t \to \infty$, and find the limiting values of $a(t)$ and $b(t)$.

f) A chemist claims that the reaction has effectively finished after 6 minutes. Using your model, comment on this claim, and state whether the system continues to move closer to equilibrium after this time.

g) Explain how the constant terms in the differential equations influence the long-term behaviour of the system.

Question 26. An ecologist is studying the interaction between two mammal species in a woodland: pine martens and red squirrels. Let $p(t)$ be the number of pine martens and $r(t)$ be the number of red squirrels at time t years. Their populations are modelled by the system $\frac{dp}{dt} = 2p - 2r - 341e^{-t}$ and $\frac{dr}{dt} = 4p - 4r$.

a) Show that p satisfies $\frac{d^2p}{dt^2} + 2\frac{dp}{dt} = -1023e^{-t}$.

b) Solve this differential equation to obtain the general solution for $p(t)$.

c) Hence find an expression for $r(t)$.

d) Initially, there are 2046 pine martens and 3411 red squirrels. Find the particular solutions for $p(t)$ and $r(t)$.

e) Determine the value of $t = T$ for which the pine marten population becomes zero. Give your answer to 3 decimal places.

f) Describe the long-term behaviour of the red squirrel population as $t \to \infty$.

Question 27. A pair of interacting quantities $x(t)$ and $y(t)$ are monitored in a system over time t. The following model is proposed:

- The rate of change of x is proportional to the difference between x and y.
- The rate of change of y is proportional to its current value but is reduced by an amount proportional to x.

This leads to the system $\dfrac{dx}{dt} = a(x - y)$ and $\dfrac{dy}{dt} = ay - bx$, where a and b are positive constants.

a) Show that x satisfies a second-order differential equation of the form $\dfrac{d^2x}{dt^2} - 2a\dfrac{dx}{dt} + a(a - b)x = 0$.

b) Hence show that the general solution for x can be written as $x(t) = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$.

c) Using your result from part (b), find the general solution for $y(t)$ in terms of A, B, a, b and t.

d) Show that there exist constants K_1 and K_2 such that $x(t) + \sqrt{\dfrac{a}{b}}y(t) = K_1e^{(a-\sqrt{ab})t}$ and $x(t) - \sqrt{\dfrac{a}{b}}y(t) = K_2e^{(a+\sqrt{ab})t}$.

e) Assuming that $K_2 \neq 0$, determine $\lim\limits_{t\to\infty} \dfrac{y(t)}{x(t)}$.

f) A scientist claims: "Regardless of the initial values of x and y, the system eventually settles into a fixed ratio between the two quantities." Discuss whether this claim is valid under the model, making your reasoning clear.

Modelling: 3 Variables

Question 28. A chemical process involves three substances: A, B, and C. Substance A is converted into B, and B is converted into C. No material is lost during each stage of the process. The amounts (in kg) at time t hours are a, b, and c, respectively. It is known that:

- A decreases at a rate proportional to its amount.
- B increases due to conversion from A, and decreases due to conversion into C.
- C is formed from B.

a) Write down differential equations for $\dfrac{da}{dt}$ and $\dfrac{db}{dt}$, introducing suitable constants.

b) Explain why a third differential equation for $\dfrac{dc}{dt}$ is not independent.

Question 29. In a production process, material P is converted into Q, and then into R. No material is lost during each stage of the process. The amounts (in kg) at time t hours are p, q, and r. The system is modelled by $\frac{dp}{dt} = -2p$ and $\frac{dq}{dt} = 2p - q$. Initially, $p = 10$, $q = 0$, and $r = 0$.

a) Show that $p = 10e^{-2t}$.

b) Show that $\frac{dp}{dt} + \frac{dq}{dt} + \frac{dr}{dt} = 0$.

c) Explain what the result in part (b) means in the context of the process.

d) Find an expression for q in terms of t.

e) Find an expression for r in terms of t.

Question 30. A substance X is processed in stages: it is converted into Y, and then into Z. No material is lost during each stage of the process. The amounts (in kg) at time t hours are x, y, and z. The system is modelled by $\frac{dx}{dt} = -\frac{3}{100}x$ and $\frac{dy}{dt} = \frac{3}{100}x - \frac{1}{100}y$. Initially, $x = 100$, $y = 0$, and $z = 0$.

a) Find x in terms of t.

b) By differentiating the equation for $\frac{dy}{dt}$ and eliminating x, show that y satisfies the second-order differential equation $\frac{d^2y}{dt^2} + \frac{1}{25}\frac{dy}{dt} + \frac{3}{10000}y = 0$.

c) Find an expression for y in terms of t.

d) Find an expression for z in terms of t, and hence show that the amount of Z present after a long time is 100 kg.